Part 20: Bayesian Estimation

Summary

To this point, our discussion has focused on the frequentist or classical approach to estimation and hypothesis testing. In that framework, consideration is given to how procedures perform in repeated use. Indeed, the performance of estimators can be assessed without using any actual data. Some find this troubling.

An alternative philosophy is the Bayesian approach to estimation. The central characteristic is that uncertainty in parameters is quantified using probability distributions.

As the name implies, Bayes' Rule plays a central role.

First, recall Bayes' Rule: If $B_1, B_2, \ldots, B_m$ are a partition of Ω, then

$$P(B_i|A) = P(A|B_i)P(B_i) \Big/ \sum_j P(A|B_j)P(B_j)$$

Also recall the continuous version:

$$f_{Y|X}(y|x) = f_{X|Y}(x|y)f_Y(y) \Big/ \int f_{X|Y}(x|u)f_Y(u)\,du$$

Here we will use this result with the parameter θ "playing the role" of Y:

$$\pi(\theta|\mathbf{x}) = f(\mathbf{x}|\theta)\pi(\theta) \Big/ \int f(\mathbf{x}|t)\pi(t)\,dt$$

The vector $\mathbf{x}$ is observed data.

The density $\pi(\theta)$ is the prior distribution for θ.

The conditional density $\pi(\theta|\mathbf{x})$ is the posterior distribution for θ.

In the Bayesian framework, the prior reflects the someone's **opinion** regarding the value of θ, **prior to** the observation of the data.

The posterior is this opinion "updated" to take the data into consideration.

How is the prior chosen? This can be tricky, and subjective.

The use of the Bayesian approach was long limited by computational difficulties associated with calculating the required integral. With the advent of Markov Chain Monte Carlo (MCMC) algorithms, these issues have largely gone away, and the use of Bayesian methods has grown greatly.

In some relatively simple cases, analytical solutions are possible.

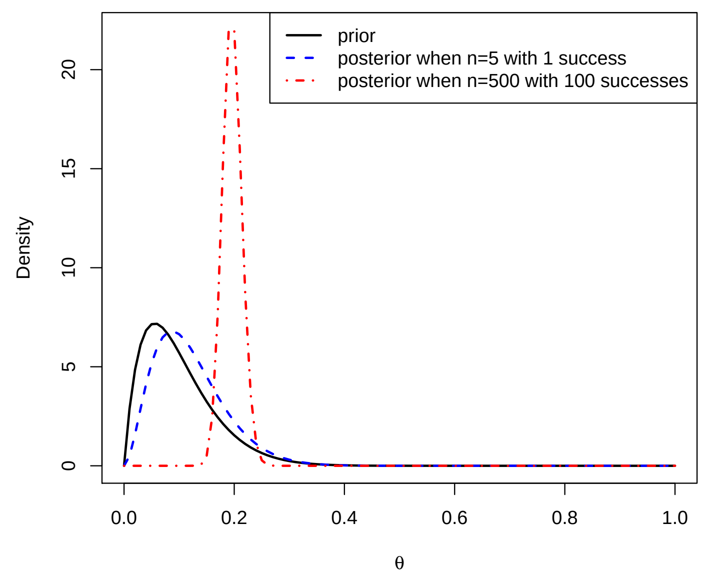
Exercise: Suppose that the observable data are modelled by i.i.d. Bernoulli(θ) random variables $X_1, X_2, \ldots, X_n$. You quantify your prior beliefs regarding θ by using the Beta(α, β) distribution, for some chosen values of α and β. Derive the posterior for θ in this case.

In this example, we see that the prior and posterior are from the same family of distributions (i.e., they are both Beta). We say that "The Beta distribution is conjugate to sampling from the Bernoulli."

Let's consider a couple specific examples. Suppose I chose $\alpha = 2$ and $\beta = 18$ to capture my prior opinion regarding θ. I observe $n = 5$ trials, and $\sum_i x_i = 1$. The result above states that the posterior is Beta$(2+1, 18+(5-1))$.

If instead, $n = 500$ and $\sum_i = 100$, the posterior is Beta$(102, 418)$.

The figures on the following slide compare these cases.



Exercise: Comment on the shapes of the posteriors in the previous example, as compared to the prior. What is the role of the sample size?

The Bayesian often wants to convert the posterior into an estimate for θ. One way to do this is to find the maximum of the posterior, called the maximum a posteriori (MAP) estimator.

In our example this comes out to be

$$\frac{\alpha + \sum x_i - 1}{\alpha + \beta + n - 4}$$

Another approach is to find the posterior mean, called the Bayes estimator.

In our example the Bayes estimator is

$$E(\theta|\mathbf{x}) = \frac{\alpha + \sum x_i}{\alpha + \beta + n} = \left(\frac{\sum x_i}{n}\right)\left(\frac{n}{\alpha + \beta + n}\right) + \left(\frac{\alpha}{\alpha + \beta}\right)\left(\frac{\alpha + \beta}{\alpha + \beta + n}\right)$$

In other words, it is a weighted average of the sample proportion and the prior mean. The weight placed on the sample proportion grows with n.

Exercise: Return to our Bernoulli example. Choose $\alpha = \beta = 1$ for the prior. What is the appeal of this choice? What is the Bayes estimator?

Exercise: The observable data are modelled using the Poisson(θ) distribution. Use the Gamma(α, β) distribution as the prior on θ. Find the posterior distribution and the Bayes estimator.

Higher Dimensional Parameters

The Bayesian approach can be extended to multi-dimensional parameters. The difficulties are more computational than conceptual. It is possible, for instance, to place prior distributions on covariance matrices with tools tailored to this purpose. Choosing priors can be even trickier in higher dimensions, however.

For example, the paper "Bayesian Portfolio Analysis" by Avramov and Zhou (*Ann. Rev. of Financial Economics*, Vol. 2) provides a nice overview of Bayesian tools for selecting portfolios. Consider Equation (8) for the underlying optimization problem. Note how uncertainty in the mean vector μ and covariance matrix V are taken into consideration: Fixed estimates are not plugged in, instead one integrates over the posterior for these quantities.